RESEARCH STATEMENT

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Recent advancements in the fields of machine learning and artificial intelligence have opened up new possibilities in the field of urban transportation systems. Machine learning algorithms like reinforcement learning and deep learning algorithms have been widely used in the fields of autonomous driving, energy-efficient driving, adaptive signal control, and vehicle routing optimization [1]. Deep learning algorithms like Convolutional Neural Networks and Support Vector Machines have been used in perception mechanisms like classification and object detection [2]. This has paved the way to design transportation systems with minimal human intervention. In light of this, my research aims to combine statistical methods with the latest deep learning technologies to design sustainable solutions for effectively managing the flow of traffic in urban systems.

I completed a bachelor of engineering degree from Tribhuvan University, Nepal in 2023. In my undergraduate studies, my coursework focused on mathematical and statistical subjects to develop a strong background for my future research. From a theoretical standpoint, I have a strong foundation in the principles of transportation engineering through coursework in transportation engineering modules. I have programming experience in Python and C, including some background in machine learning algorithms through online courses and training. I aim to generate effective solutions for the problems faced in the transportation engineering ecosystem.

For my graduate studies, I have two prospective research projects in mind. The projects are based on machine learning and deep learning algorithms to design effective urban mobility solutions. The first one is an **adaptive signal control for traffic lights**. Traditional traffic signal control suffers from scalability issues while not being responsive to traffic congestion and emergency situations. Modern deep learning methods like Deep Q Learning (DQL) and Deep Deterministic Policy Gradient (DDPG) are proving efficient, providing scalable solutions to traffic light problems in a single road intersection as well as a grid network [3]. These methods, however, lack generalization and interpretability. With the rapid rise of large language models, these LLM agents are emerging as a solution to interpretability issues. LLMs can be taught to conduct human-like, step-wise analysis of traffic conditions. This would tempt the agent to choose the optimal control action, enhancing overall efficiency at the intersection [4]. However, LLMs suffer from other issues like a lack of quantitative optimization, high computational requirements, and hallucination risk. LLMs cannot perform the numerical optimization required for signal timing, may encounter hallucination issues, thereby incorrectly interpreting traffic conditions, and have high computation costs for real-time traffic control. A new innovative approach combining both Deep Reinforcement Learning and LLMs can provide an optimal solution for signal control. DRL can be used for signal timing, whereas LLMs can be used for processing contextual information like events and constructions. The insights from LLMs can be

used by DRL to make final control decisions. This would solve the issue of using each of them individually while being efficient and scalable.

The second idea is to work on **autonomous vehicle risk assessment**. With the increase in computational power, rapid use of CNNs is seen in the technical community, with more focus on making vehicles fully autonomous. Most people in today's world still have safety concerns over the use of self-driving cars. With NHTSA reporting over 1,000 accidents for Tesla Cars after autopilot was employed, people's confidence in using autonomous vehicles seems reportedly low [5]. Since the real-world driving environment is different and the performance of self-driving cars depends on environmental conditions, developing risk assessment algorithms to identify risk in real time is required [6]. The algorithm should evaluate existing conditions to determine the probability of a potential risk. This could be a fun project to work on.

I hope to work with the best faculty members and industry professionals throughout my graduate studies to find potential solutions to the problems outlined above.

Abbreviations used in the text*: DRL= Deep Reinforcement Learning, CNNs= Convolutional Neural Networks, NHTSA= National Highway Traffic Safety Administration, LLMs=Large Language Models*

References

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